

CONFIDENCE INTERVALS

master recipe • also used in Units 4 and 5

ZONE A • THE FORMULAS

$$\text{statistic} \pm \underbrace{(\text{critical value})(\text{standard error})}_{\text{margin of error}}$$

$$ME = z^* \cdot SE \quad (\text{proportions})$$

$$ME = t^* \cdot SE \quad (\text{means})$$

Use the procedure's df to choose t^* .

$$\text{width} \propto \frac{1}{\sqrt{n}} \Rightarrow \text{halve the margin} \Rightarrow \text{quadruple } n$$

THE FOUR-STEP FRAME

- 1 STATE** Name the parameter in context.
- 2 PLAN** Name the interval; check conditions.
- 3 DO** Formula, substitutions, and interval.
- 4 CONCLUDE** Interpret the interval in context.

COMMON z^* VALUES

confidence	z^*
90%	1.645
95%	1.960
99%	2.576

Get t^* from the table or calculator using **df**.

ZONE B • CONDITIONS — ALWAYS IN THIS ORDER

1 • RANDOM

Random sample or
randomized experiment.

2 • 10%

Sampling without replacement:
 $n < 0.10N$.

3 • NORMAL / LARGE COUNTS

CI: $n\hat{p} \geq 10$, $n(1 - \hat{p}) \geq 10$

Test: $np_0 \geq 10$, $n(1 - p_0) \geq 10$ (see P11)

means: normal population, $n \geq 30$, or no strong
skew/outliers

ZONE C • SAY IT IN WORDS

"We are _____% confident that the interval from _____ to _____ captures the true _____."

Confidence level: If we took many samples and built an interval each time,
about _____% of those intervals would capture the true _____.

ZONE D • WATCH OUT

THE INTERVAL ESTIMATES THE PARAMETER. THE LEVEL DESCRIBES THE METHOD.

Never say the fixed parameter has a 95% chance of being inside the interval. After the interval is calculated,
it either captures the parameter or it does not.